

MedQuAD Medical Q&A Chatbot

End-to-End Project Report

Project: Medical Question Answering Chatbot

Dataset: MedQuAD

Interface: Streamlit

Retrieval: Semantic + Keyword Hybrid Retrieval

1. Project Overview

This project implements a Medical Question Answering chatbot using the MedQuAD dataset. A user enters a medical question and the system retrieves the most relevant answer from the prepared knowledge base. Sentence Transformers provide semantic representations, FAISS performs vector similarity search, and keyword matching strengthens retrieval for important medical terms.

2. Project Objective

The objective is to develop a retrieval-based Medical Q&A chatbot using MedQuAD. The project covers data preparation, question embeddings, FAISS indexing, hybrid retrieval, result ranking and answer presentation through Streamlit.

3. Dataset

MedQuAD is used as the medical knowledge base. It contains medical question-answer pairs collected from medical information sources. The project organizes questions, answers and related metadata for retrieval.

4. Data Preprocessing

The dataset is cleaned and structured. Metadata includes question ID, focus, question type, synonyms, source, source file and URL. The processed metadata is stored in `qa_metadata.pkl`.

5. Sentence Transformer Embeddings

The project uses sentence-transformers/all-MiniLM-L6-v2. Each stored question is converted into an embedding representing its semantic meaning. The user's question is converted using the same model.

6. FAISS Index

The question embeddings are stored in `medical_qa.index`. During a query, FAISS retrieves a candidate set of questions with high semantic similarity.

7. Hybrid Retrieval

The retriever combines semantic similarity with lexical keyword matching. Query tokens are cleaned using stop words, and IDF weighting gives more importance to less common terms. Exact phrase and medical-term bonuses are also applied.

8. Hybrid Scoring

The primary score combines 60 percent semantic similarity and 40 percent lexical similarity, with additional phrase and medical-term bonuses. Results are sorted by the final score and duplicate questions are removed.

9. Streamlit Application

The Streamlit application is the user interface. A user enters a question and clicks Search. The application calls MedicalRetriever, retrieves the best MedQuAD result, and displays the answer and source. Session state allows multiple questions to remain visible.

10. End-to-End Workflow

The complete flow is: user question → text processing and embedding → FAISS semantic search → keyword candidate search → candidate combination → hybrid scoring → ranking → best MedQuAD answer → Streamlit display.

11. Project Structure

app.py contains the Streamlit interface. src/medical_retriever.py contains the hybrid retrieval logic. models/medical_qa.index contains the FAISS vector index. models/qa_metadata.pkl contains questions, answers and metadata.

12. Running the Project

Activate the Python environment, install requirements.txt, and run `python -m streamlit run app.py` from the project root. Streamlit normally opens the application at <http://localhost:8501>.

13. Testing

Example test questions include: What are the symptoms of influenza? What causes diabetes? What are the symptoms of arthritis? What is insulin used for? Testing verifies that questions are accepted and relevant MedQuAD answers are returned.

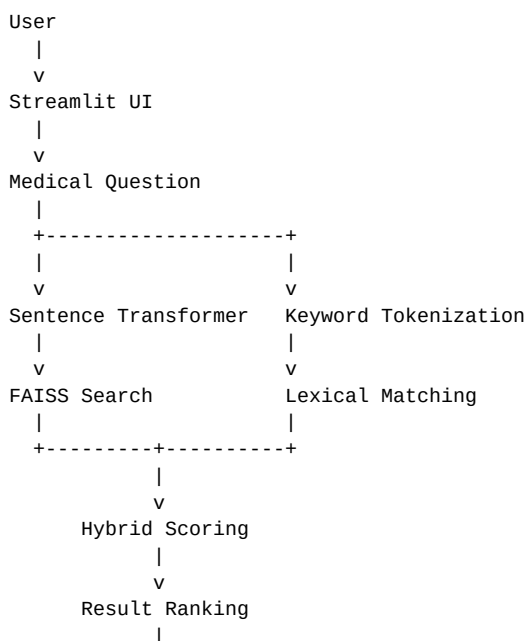
14. Limitations

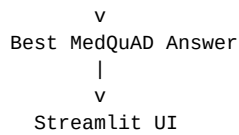
The chatbot is limited to information available in the MedQuAD knowledge base. It does not independently diagnose conditions or generate new medical knowledge. Retrieval quality depends on dataset coverage and similarity between the user's question and stored questions.

15. Conclusion

The project demonstrates an end-to-end Medical Q&A; chatbot using MedQuAD. Combining Sentence Transformers, FAISS semantic retrieval and keyword matching provides a practical retrieval workflow, while Streamlit provides a simple interface for asking multiple medical questions.

System Architecture





Project Deliverables

Component	Purpose
MedQuAD dataset	Medical Q&A knowledge base
qa_metadata.pkl	Questions, answers and metadata
medical_qa.index	FAISS vector index
medical_retriever.py	Semantic + keyword hybrid retrieval
app.py	Streamlit chatbot interface
README.md	Project documentation

Disclaimer: This project is for educational and informational purposes only and should not be used as a substitute for professional medical advice, diagnosis or treatment.